

[T]-Theory: Social Science

P11 · Collective Field · For: The Sociologist

The Rapport Propagator — social interaction as wave phenomenon.

Social field phenomena:

- Rapport = field coupling $J_A \cdot G \cdot J_B$ between two agents
- Cultural contagion = wave propagation in the social Green's function
- Dialect spread = acoustic field diffusing over a geographic network
- Social norms = stable attractor basins in collective BRECVEMA space

Scale 9 identification:

- Scale 9 = society/city: Social Interaction Kernel
- Coordination = frequency locking between individual fields
- Institutions = high- k (short-range) field configurations
- Revolutions = $k \rightarrow 0$ (long-range correlation onset)

I · The Rapport Propagator

Rapport between agents A and B is the two-point function:

$$\langle \Psi_A(t) \Psi_B(t') \rangle = G_{\text{rapport}}(t - t')$$

The correlation decays as $e^{-|t-t'|/\tau_{\text{social}}}$ where τ_{social} is the social memory timescale.

Strong rapport = low k_9 (long correlation length in social space). Weak rapport = high k_9 (rapid decorrelation). Collective intelligence = correlated field achieving $k_9 \rightarrow 0$.

II · Structural Contagion

Emotional contagion in cities (dialect spread, fashion, opinion):

$$\frac{\partial \Psi}{\partial t} = D \nabla^2 \Psi - k^2 \Psi + J_{\text{media}}(x, t)$$

This is the **diffusion-reaction equation** for social fields. The wavenumber k_9 sets the social correlation length $\ell = 1/k_9$. At scale 9, ℓ ranges from a city block (local dialect) to a continent (global cultural attractor).

III · Institutions as Field Configurations

- **High- k institutions:** tight norms, short-range influence (local customs)
- **Low- k institutions:** broad norms, long-range reach (global law, internet)
- **Phase transitions:** social revolutions = $k \rightarrow 0$ criticality in the social field
- **Collective memory:** the social memory kernel $K(\tau) =$ institutional precedent

IV · The Co-regulation Model

Mathematical co-identification (P3): social co-regulation $\equiv$ field coupling:

$$\dot{\Psi}_A = -\nabla H_A + \alpha G \cdot (\Psi_B - \Psi_A) + \eta_A$$

The co-regulation coupling constant α is the sociological equivalent of the electromagnetic fine-structure constant. High α = cohesive societies; low α = atomised societies.

G-ID: *The Rapport Propagator* — social interaction kernel for coordination and contagion waves. ORCID: 0009-0007-2194-0850
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